Group Coursework Submission Form

Specialist Masters Programme

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Lecturer: Rosalba Radice	**Submission Date:** December 6, 2024

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Table of contents

1. Explanation R Code 1 ..... *2*

1.1 Functionality of the Code **2**

1.2 Commentary on the Output **3**

1.3 Conclusion **4**

2. Time Series Analysis *5*

2.1 Time Series 1 Analysis: **5**

2.2 Time Series 2 Analysis: **9**

2.3 Time Series 3 Analysis: **12**

3. Explanation R Code 2 ..... *15*

3.1 Ordinary Nonparametric Bootstrap Analysis **15**

3.2 Block Bootstrap Analysis: **16**

3.3 Comparative Analysis and Conclusion: **17**

4. Appendix *18*

4.1 R Code Question 1 ..... **18**

4.2 R Code Question 2 ..... **18**

4.3 R Code Question 3 ..... **22**

1. Explanation R Code 1

1.1 Functionality of the Code

This R code uses the **boot** package to perform a **non-parametric bootstrap**. The goal is to estimate the **95th percentile** of a sample drawn from an **exponential distribution**. Here is a step-by-step breakdown of the process:

1. **Loading the boot library**: The **boot** library provides functions for **bootstrap resampling**, which involves repeatedly sampling with replacement from the data to estimate properties of the underlying distribution.
2. **Generating Data**: A dataset of size **n = 20** is generated from an **exponential distribution** with a rate parameter ($\lambda = 1 \backslash lambda = 1\lambda = 1$). This is achieved using the R function **rexp()**, which simulates random numbers from an exponential distribution.
3. **Defining the Statistic Function**: A custom function, **quantile.stat**, is defined. This function takes two arguments: the dataset (**data**) and a vector of **resampling indices**. The function calculates the **95th percentile** of the resampled dataset using the **quantile ()** function.
4. **Bootstrapping**: The **boot()** function is applied to perform **R = 1000** bootstrap resamples. It estimates the sampling distribution of the **95th percentile** by repeatedly computing it for each resample.
5. **Output and True Value Comparison**:
    a. The bootstrap results are printed to display statistics such as the **original estimate**, **bias**, and **standard error**.
    b. A plot visualizes the distribution of the bootstrap estimates.
    c. The **true 95th percentile** of the exponential distribution is analytically computed using **qexp()**.

Bootstrap Statistics:

Method	original	std.error	bias
Non-Parametric Bootstrap	2.791975	1.051627	-0.2121646

Figure 1: Histogram of t

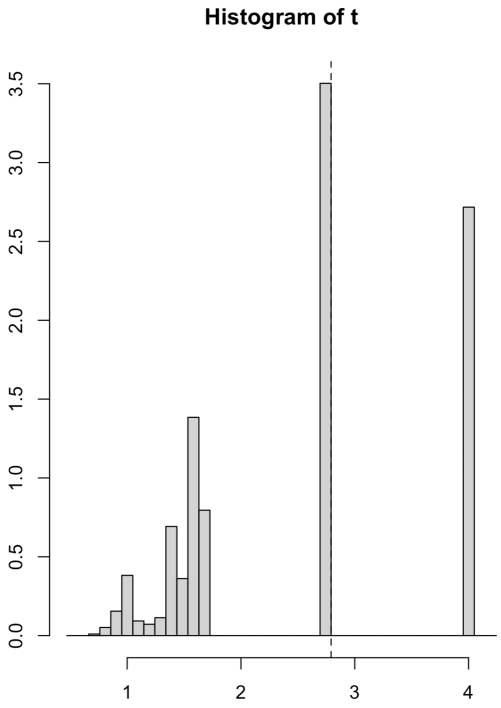
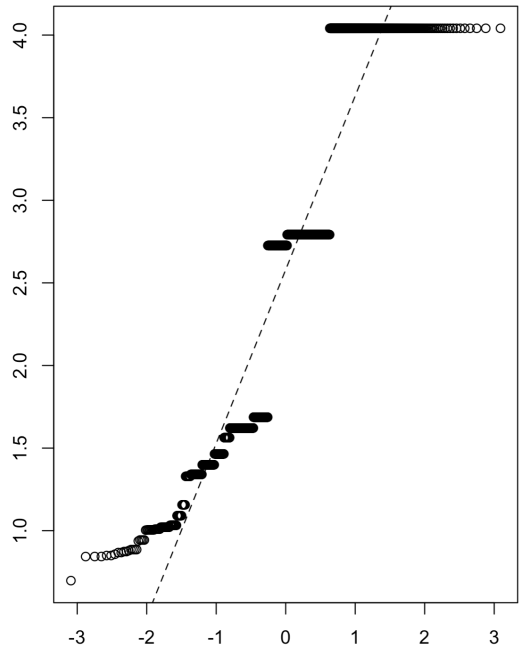


Figure 2: Quantiles of Standard Normal



1.2 Commentary on the Output

- **Bootstrap Results**:

- The **bootstrap estimate** for the 95th percentile (original statistic) is approximately **2.792**.
- The **bias** of the estimate (difference between the bootstrap mean and original statistic) is about **-0.212**, indicating a slight underestimation.
- The **standard error** (variability of the bootstrap estimates) is **1.052**, reflecting the uncertainty due to resampling.

- **True 95th Percentile**:

- The true value of the 95th percentile for the exponential distribution is calculated as approximately **2.996** using **qexp()**.

- **Comparison and Interpretation**:

- The bootstrap mean (approximately **2.80**) slightly underestimates the true 95th percentile (**2.996**). This discrepancy arises due to the **small sample size (n = 20)**, leading to increased variability and bias in the bootstrap estimates.
- The **standard deviation (1.06)** of the bootstrap estimates represents the sample's variability and reflects the uncertainty of the percentile estimation.

➢ **Plot Insights**:

- The plot of the bootstrap results shows the **distribution of the 95th percentile estimates** across 1000 resamples.
- Given the skewness of the exponential distribution, the plot exhibits a **positively skewed distribution**, with most estimates clustered near the true percentile and a tail extending toward higher values.

1.3 Conclusion

This code demonstrates how **non-parametric bootstrapping** can estimate the 95th percentile of a dataset sampled from an exponential distribution. Key takeaways include:

1. **Strengths**:
    a. Bootstrapping provides a flexible, distribution-free approach to estimate statistics like percentiles without relying on strict parametric assumptions.
    b. It is particularly useful for small samples where traditional methods may not be robust.
2. **Limitations**:
    a. With a small sample size (*n=20)*, the bootstrap estimates are more variable and biased, as evidenced by underestimation of the true value and the high standard error.
    b. Skewed distributions, such as the exponential distribution, can exacerbate these issues, requiring careful interpretation.
3. **Recommendations**:
    a. Increasing the sample size would reduce the bias and variability in bootstrap estimates.
    b. For highly skewed distributions, combining bootstrap results with other analytical or parametric methods may yield more accurate results.

2. Time Series Analysis

Figure 3a: Time Series 1

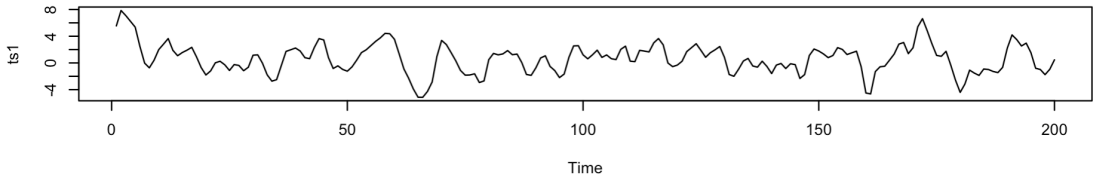


Figure 3b:Time Series 2

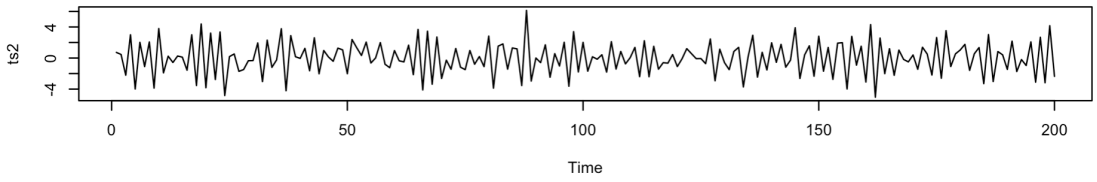
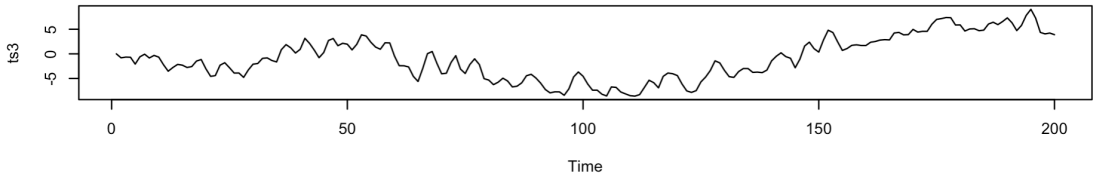


Figure 3c:Time Series 3



2.1 Time Series 1 Analysis:

Upon conducting a detailed analysis of the ***Figure 3a*** graph of the ts1 series, it can be observed that the time series lacks a clear long-term trend, as the values oscillate around zero over time. Regarding short-term trends or seasonal components, a slight pattern is identified, but it does not exhibit fixed periodicity, and peaks occur at random points in time. Additionally, the variance remains generally constant, although it increases randomly during certain periods without definitively altering the overall pattern.

Based on this analysis, it can be concluded that the series may not require differentiation, or at most, it may need a first-order differentiation. This will be validated through the analysis of the residuals and the AIC criterion to identify the best-fitting model.

Figure 4a: ACF TS1

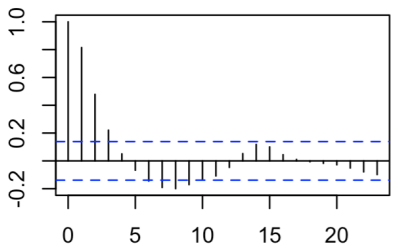


Figure 4b: PACF TS1

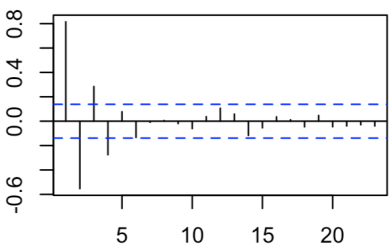


Figure 4c: ACF TS1.Diff(1)

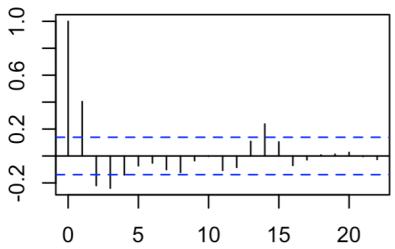
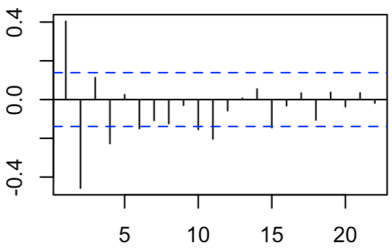


Figure 4d: PACF TS1.Diff(1)



According to the ***Figure 4c and 4d*** ACF and PACF graphs, the first-order differentiation significantly improves the analysis of the series by partially correcting the pattern observed in the original series and accelerating the decay in the ACF. However, this differentiation does not completely resolve the issue, as significant values are still observed at periods between 5 and 20. This may indicate unidentified seasonality, which is common in real-world time series where decay patterns do not perfectly align with theoretical expectations.

The ACF of the differentiated series reveals a rapid decay at lag 3, suggesting that the number of moving average terms (q) in the ARIMA model could be 3. Similarly, the PACF indicates that the first two lags are significant before starting to decay, implying that the number of autoregressive terms (p) could be 2. Based on this analysis, the **ARIMA (2,1,3)** model is proposed as the most suitable to capture the series dynamics. Additionally, to ensure a parsimonious approach, the **ARIMA (2,1,2)** model will also be evaluated, aiming to balance simplicity and model fit.

Figure 5: Residual Analysis for ARIMA(2,1,3) Model

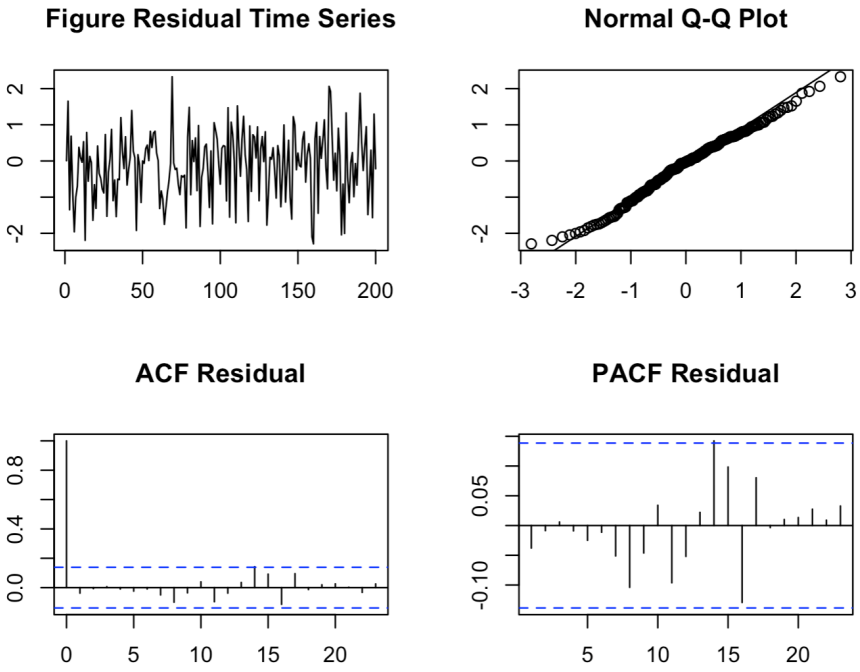
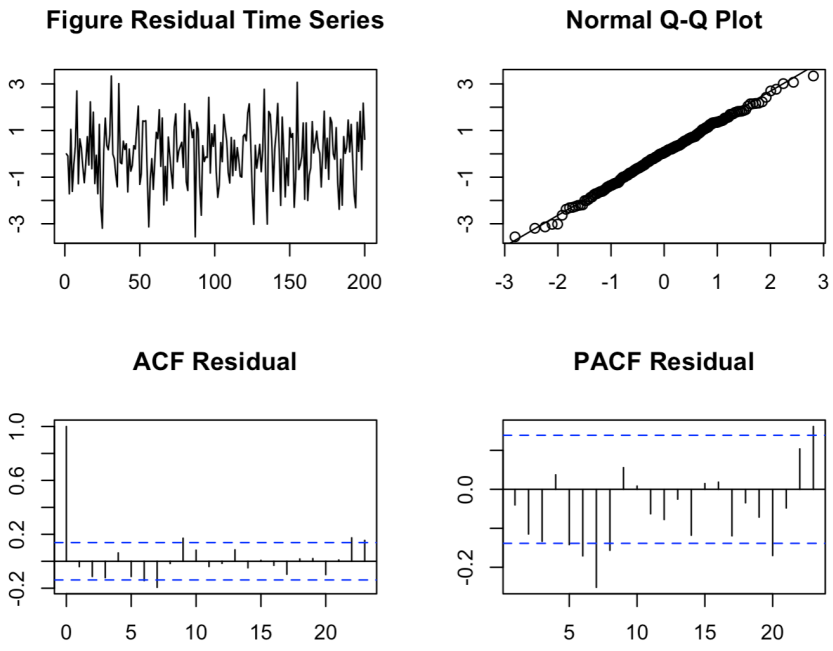


Figure 6: Residual Analysis for ARIMA(2,1,2) Model



The residuals of both models exhibit a constant mean and variance around zero. In the Q-Q plots, both models adequately fit a normal distribution. However, while **Model 1: ARIMA (2,1,3)** meets the assumptions that the residuals behave as Gaussian white noise, **Model 2: ARIMA (2,1,2)**, although simpler, does not fully satisfy all the requirements due to the presence of some significant points. This suggests that its residuals may not originate from a white noise process, indicating potential inconsistencies with the model assumptions. Nevertheless, since we are dealing with a real-world time series and these deviations can be considered acceptable, we decide to utilize the AIC criterion.

Table 1: AIC Values for Different ARIMA Models

Model	AIC
ARIMA(2,1,3)	556.3270
ARIMA(2,1,2)	695.1681

Source: Self-prepared based on model outputs.

This criterion showed in ***Table 1*** indicates that the model with the lowest value is **Model 1**, and therefore, this model is selected for the final analysis.

$$\boldsymbol{X_t - X_{t-1}} = \alpha_1\left(X_{(t-1)} - X_{t-2}\right) + \alpha_2(X_{t-2} - X_{t-3}) + \beta_1 E_{(t-1)} + \beta_2 E_{t-2} + \beta_3 E_{t-3} + E_t + C$$

Table 2: ARIMA (2,1,3) Model Results

Section	Name	Value	Std_Error
Parameters	ar1	0.7004	0.5320
	ar2	-0.0993	0.3871
	ma1	-0.0935	0.5358
	ma2	-0.7438	0.1160
	ma3	-0.1628	0.4326
Model Metrics	AIC	556.3300	
	BIC	576.0900	
	Log-Likelihood	-272.1600	

Source: Self-prepared based on ARIMA (2,1,3) model outputs.

2.2 Time Series 2 Analysis:

The **ts2** series, showed in ***Figure 3b*** does not show a clear long-term trend, as the values fluctuate around an approximately constant mean close to 0. Similarly, the variance remains generally constant over time.

The seasonal component is not evident and does not show specific periodic patterns, suggesting the absence of clear seasonality. Therefore, the same procedure used for the **ts1** series was applied: analysing the ACF and PACF graphs of the series without differentiation and with a first-order differentiation.

Figure 7a: ACF TS2

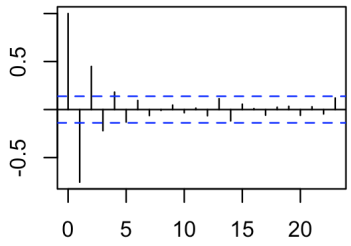


Figure 7b: PACF TS2

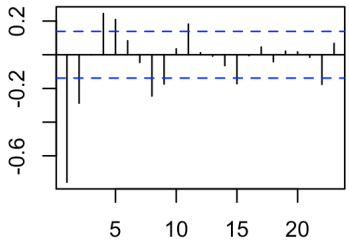


Figure 7c: ACF TS2.Diff(1)

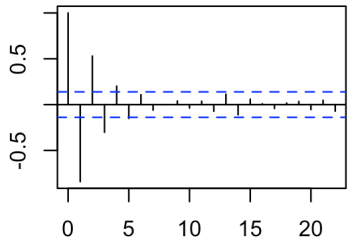
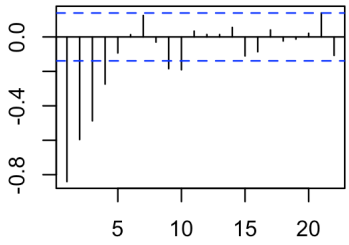


Figure 7d: PACF TS2.Diff(1)



From this analysis, we can deduce that although the first-order differentiation does not significantly improve the model analysis, two models can be proposed based on the undifferentiated series: Model 1: **ARIMA (2,0,3)** and a simpler Model 2: **ARIMA (2,0,2).** This selection is based on the significant points observed in both the ACF and PACF graphs.

Figure 8: Residual Analysis for ARIMA(2,0,3) Model

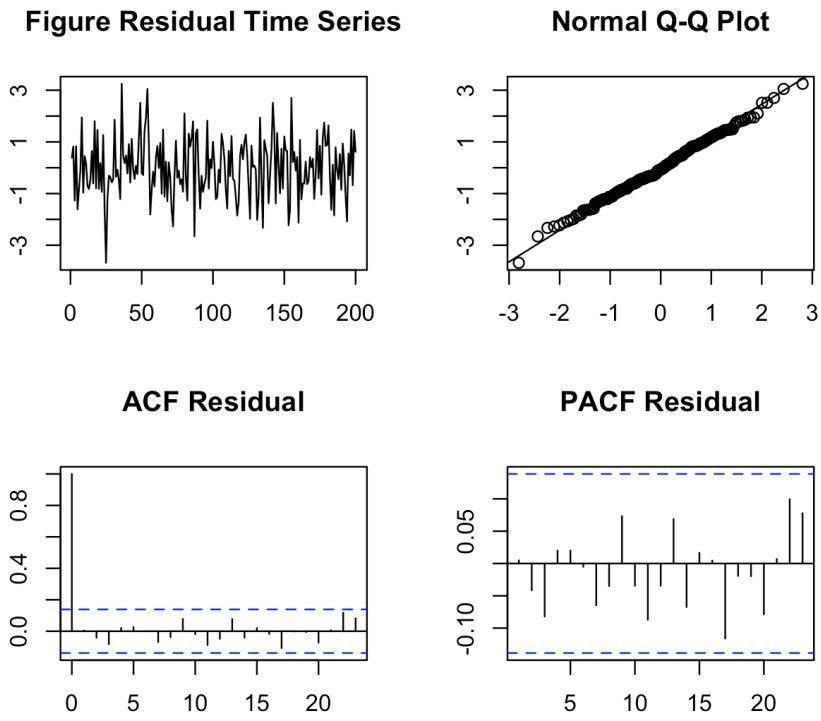
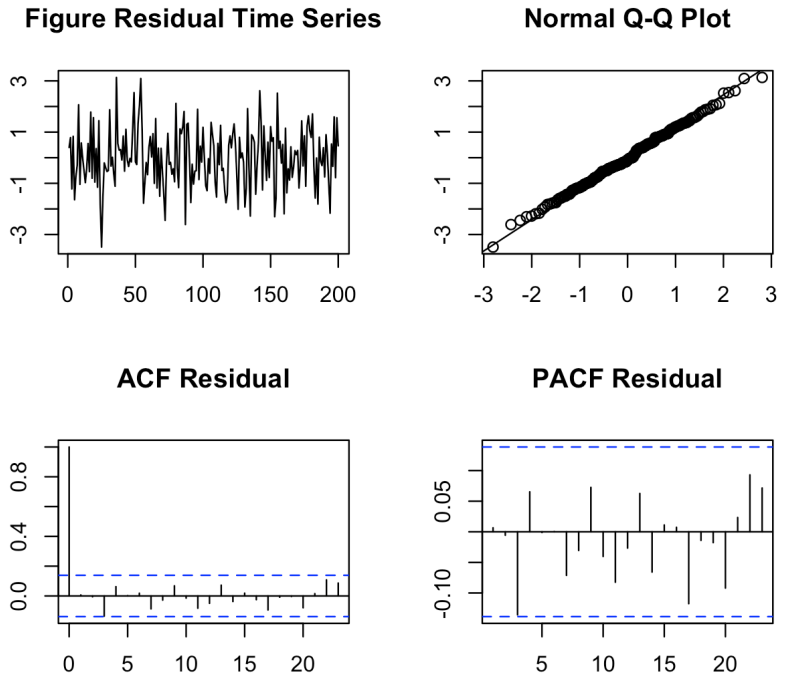


Figure 9: Residual Analysis for ARIMA(2,0,2) Model



For both models, all residual assumptions are generally satisfied (constant mean and variance around 0, normal distribution and an ACF and PACF with no significant lags beyond order 1).

Table 3: AIC Values for Different ARIMA Models

Model	AIC
ARIMA(2,0,3)	640.7399
ARIMA(2,0,2)	641.3156

Source: Self-prepared based on model outputs.

However, the AIC criterion indicates that the model with the best fit, albeit by a very small margin, is **Model 1: ARIMA (2,0,3).**

$$\boldsymbol{X_t} = \alpha_1 X_{(t-1)} + \alpha_2 X_{t-2} + \beta_1 E_{(t-1)} + \beta_2 E_{t-2} + \beta_3 E_{t-3} + E_t + C$$

Table 4: ARIMA (2,0,3) Model Results

Section	Name	Value	Std_Error
Parameters	ar1	-0.6646	0.2254
	ar2	0.0789	0.0811
	ma1	-0.4994	0.2170
	ma2	0.1895	0.2450
	ma3	0.6096	0.2025
	intercept	-0.0172	0.0661
Model Metrics	AIC	640.7400	
	BIC	663.8300	
	Log-Likelihood	-313.3700	

Source: Self-prepared based on ARIMA (2,0,3) model outputs.

2.3 Time Series 3 Analysis:

Regarding series 3, we can observe from ***Figure 3c*** a long-term trend, which experienced a break point around period 100. Before this point, the trend was decreasing, but after it, the trend started to rise. No significant change in variance is observed, so no transformation is required. Additionally, there is no visible pattern suggesting the presence of a seasonal component. To remove the trend, the first step will be to apply a first-order differentiation to analyse if this eliminates it or if an additional transformation will be necessary.

Figure 10a: ACF TS3

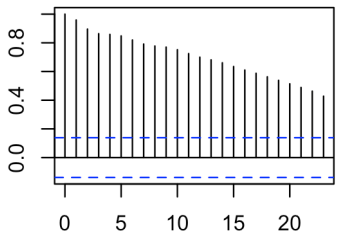


Figure 10b: PACF TS3

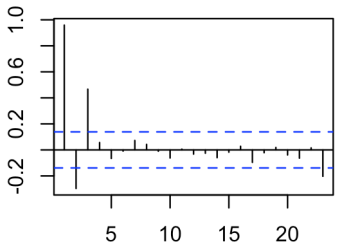


Figure 10c: ACF TS3.Diff(1)

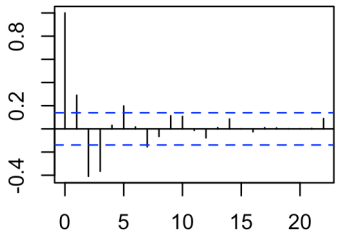
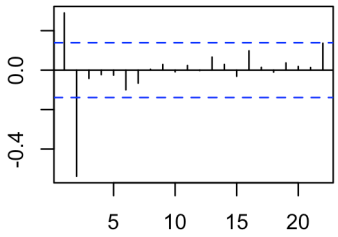


Figure 10d: PACF TS3.Diff(1)



The first-order differentiation significantly corrects the trend issue, causing the significant values in the ACF and PACF graphs to diminish rapidly.

Based on the graphs obtained after applying the first-order differentiation, two models are proposed for analysis: an **ARIMA (2,1,3)** and, as in the previous analyses, a simpler model, the **ARIMA (2,1,2)**.

Figure 11: Residual Analysis for ARIMA(2,1,3) Model

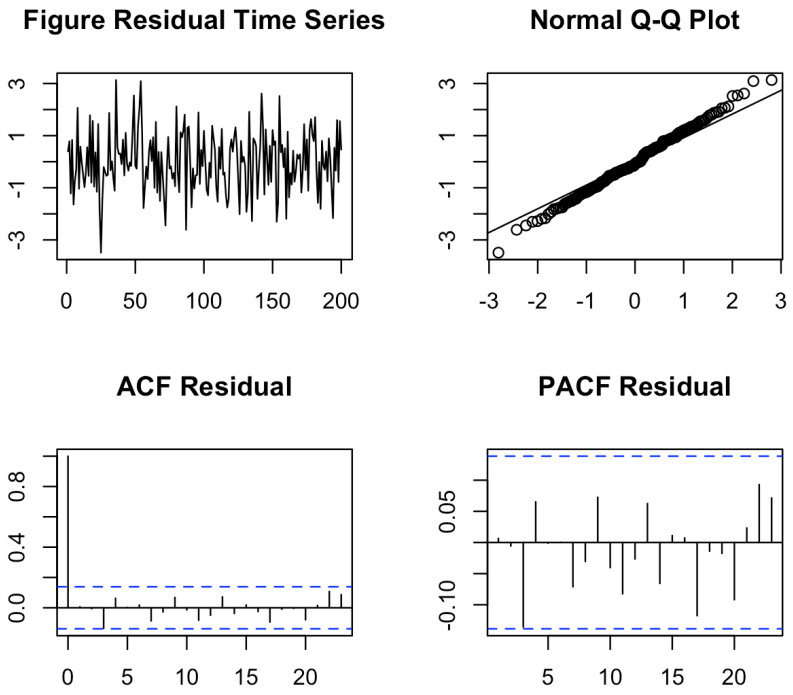
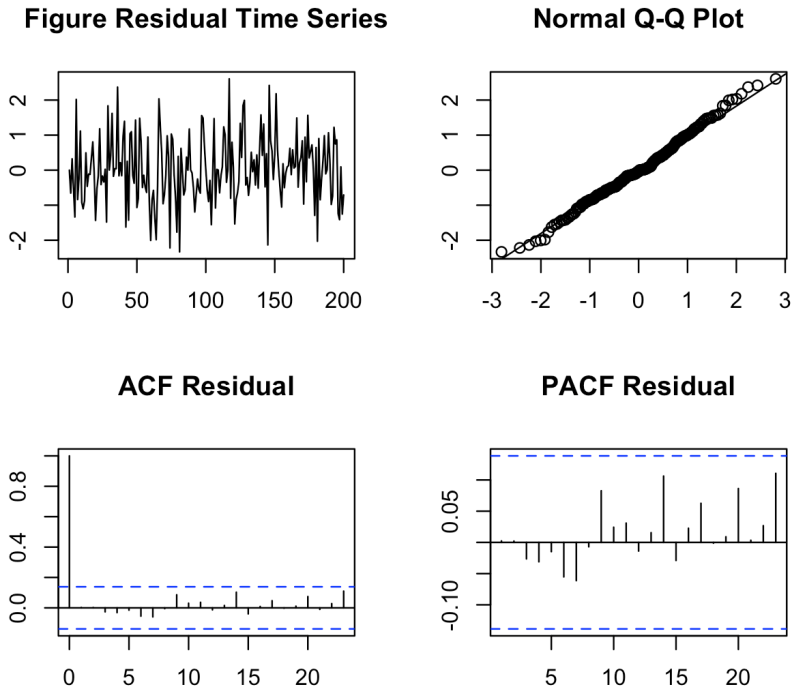


Figure 12: Residual Analysis for ARIMA(2,1,2) Model



For both models, all residual assumptions are generally satisfied (constant mean and variance around 0, normal distribution) and an ACF and PACF with no significant lags beyond order 1. However, the AIC criterion indicates that the model with the best fit is **Model 2: ARIMA (2,1,2).**

Table 5: AIC Values for Different ARIMA Models

Model	AIC
ARIMA(2,1,3)	560.1797
ARIMA(2,1,2)	558.6605

Source: Self-prepared based on model outputs.

However, the AIC criterion indicates that the model with the best fit is **Model 2: ARIMA (2,1,2).**

$$\boldsymbol{X_t - X_{t-1}} = \alpha_1\left(X_{(t-1)} - X_{t-2}\right) + \alpha_2(X_{t-2} - X_{t-3}) + \beta_1 E_{(t-1)} + \beta_2 E_{t-2} + E_t + C$$

Table 6: ARIMA (2,1,2) Model Results

Section	Name	Value	Std_Error
Parameters	ar1	0.5320	0.1348
	ar2	-0.5112	0.1170
	ma1	-0.1122	0.1571
	ma2	-0.0736	0.1749
Model Metrics	AIC	558.6600	
	BIC	575.1300	
	Log-Likelihood	-274.0900	

Source: Self-prepared based on ARIMA (2,1,2) model outputs.

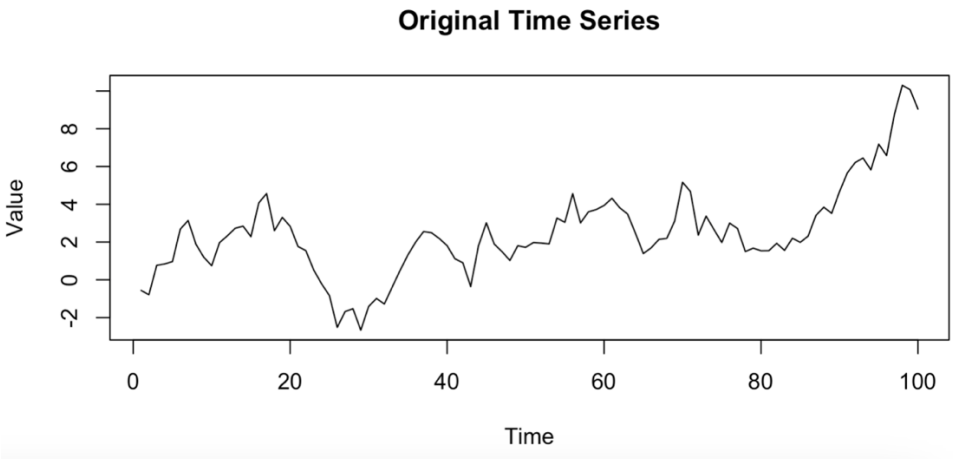
3. Explanation R Code 2

In this analysis, we demonstrate the bootstrapping techniques to estimate the mean of a time series and compares the variability of estimates under two different resampling methods i.e. Ordinal non-parametric bootstrap and block bootstrap.

The library *boot* is loaded to perform bootstrap resampling. A random seed i.e. *set.seed(123)* ensures that the results are consistent. A time series of length 100 is generated using cumulative sum of random normal values i.e. *rnorm(n)*.

This time series is visualized using the *plot()* function, as shown in the ***Figure 13***. This plot shows its cumulative trend, which demonstrates the sequential nature of the data points and their interdependencies.

Figure 13: Original time series



3.1 Ordinary Nonparametric Bootstrap Analysis

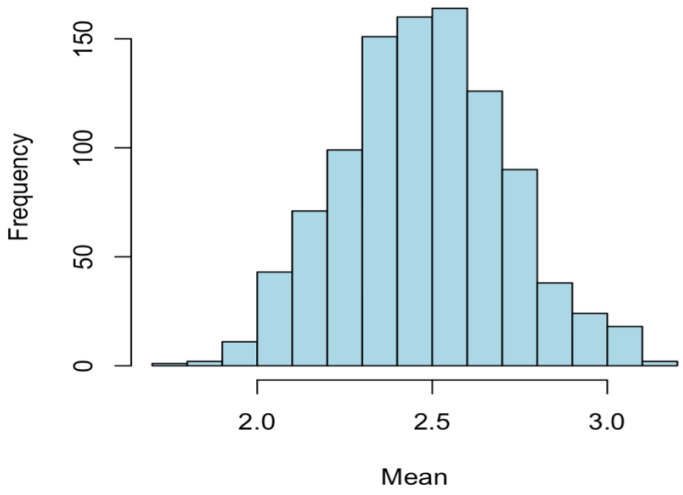
A function *mean.function* is defined to calculate the mean of resampled data. The boot function performs 1000 resamples of the time series, calculating the mean for each resample. This method assumes independence of data points, which may not hold for time series data. The ordinary bootstrap result shows an original mean of 2.473137, bias of 0.004056751, and a standard error of 0.2341277 as showed in ***Table 7***.

Table 7: Ordinary Nonparametric Bootstrap statistics

Method	Original	Bias	Std_Error
Ordinary Nonparametric Bootstrap	2.473137	0.0040568	0.2341277

The histogram of bootstrap means is symmetric and centred around the mean as shown in ***Figure 14***, which effectively captures the variability of mean. However, this method assumes independence between data points, which may not hold for time series data, potentially underestimating the true variability in the series. The relatively small bias indicates good accuracy in the point estimate, but the standard error might be misleading.

Figure 14: Ordinary Nonparametric Bootstrap



3.2 Block Bootstrap Analysis:

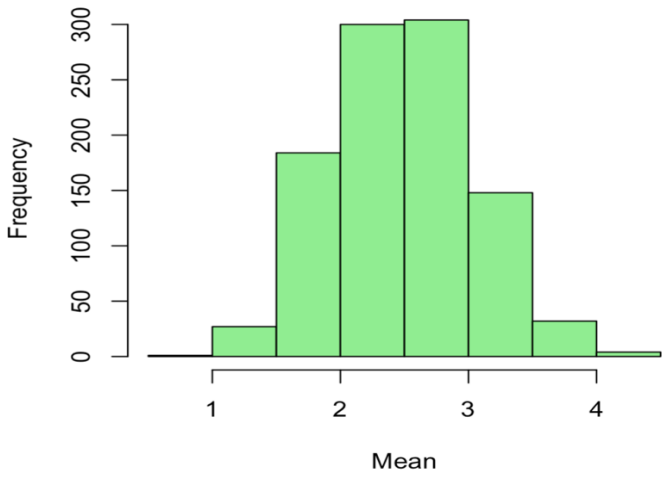
The *tsboot* function performs block bootstrap with a block size of 10 with preserving temporal dependencies. The function *mean.function.tsboot* calculates the mean of the data in each block. The block bootstrap results show the same original mean but a slightly higher bias of 0.004913312 and a larger standard error of 0.5552648 as shown in ***Table 8***.

Table 8: Block Bootstrap statistics

Method	Original	Bias	Std_Error
Block Bootstrap for Time Series	2.473137	0.0049133	0.5552648

The histogram of the blocked bootstrap mean is wider, which reflects the increased variability due to the block structure. The Block bootstrap accounts for dependencies but requires careful block size selection. Small blocks fail to capture long-range dependencies, while large blocks reduce the effective sample size and may overestimate variability. The larger standard error compared to ordinary bootstrap suggests that accounting for temporal dependence reveals greater uncertainty in the estimates.

Figure 15: Block Bootstrap



3.3 Comparative Analysis and Conclusion:

The ordinary nonparametric bootstrap is computationally efficient and simpler but assumes data points are independent which often can be invalid for a time series. This can be led to underestimation of variability and misleading confidence intervals. The block bootstrap method preserves temporal dependencies which provides more realistic assessment when careful block size is selected.

The comparison highlights the significance of choosing the appropriate method for time series data. The ordinary bootstrap is more suitable for independent data whereas block bootstrap is more appropriate for the timeseries due to its ability to account temporal dependencies. The larger standard error in the block bootstrap emphasizes to consider the time series structure when making statistical inferences.

4. Appendix

4.1 R Code Question 1

```r
library(boot)
set.seed(123)
n <- 20
data <- rexp(n, rate = 1)
quantile.stat <- function(data, indices) {
  return(quantile(data[indices], 0.95))
}
results <- boot(data, statistic = quantile.stat, R = 1000)
print(results)
plot(results)
true.95th <- qexp(0.95, rate = 1)
cat("True 95th Percentile:", true.95th, "\n")
```

4.2 R Code Question 2

```r
if (!requireNamespace("flextable", quietly = TRUE)) install.packages("flextable")
if (!requireNamespace("magrittr", quietly = TRUE)) install.packages("magrittr")
library(flextable)
library(magrittr)

data <- read.table("ARIMAsim2.dat",header = TRUE)

#-----------------------
#TS1
#-----------------------

#Data
ts1 <- ts(data["ts1"])

# Differentiation
ts1.diff <- (diff(ts1,lag = 1))
plot(ts1.diff)

par(mfrow = c(2, 2), oma = c(3, 6, 3, 6), mar = c(3, 3, 4, 2))
acf(ts1,main = "Figure 4a: ACF TS1")
pacf(ts1,main = "Figure 4b: PACF TS1")
acf(ts1.diff,main = "Figure 4c: ACF TS1.Diff(1)")
pacf(ts1.diff,main = "Figure 4d: PACF TS1.Diff(1)")

# Model and residuals
m1 <- arima(ts1, order = c(2, 1, 3))
plot(resid(m1),main="Figure Residual Time Series")
qqnorm(resid(m1)); qqline(resid(m1))
acf(resid(m1),main="ACF Residual"); pacf(resid(m1),main="PACF Residual")
mtext("Figure 5: Residual Analysis for ARIMA(2,1,3) Model", side = 3, line = 1, outer = TRUE, cex = 1)
m2 <- arima(ts2, order = c(1, 1, 2))
plot(resid(m2),main="Figure Residual Time Series")
qqnorm(resid(m2)); qqline(resid(m2))
acf(resid(m2),main="ACF Residual"); pacf(resid(m2),main="PACF Residual")
mtext("Figure 6: Residual Analysis for ARIMA(2,1,2) Model", side = 3, line = 1, outer = TRUE, cex = 1)

```

```r
AIC(m1,m2)

aic_results <- data.frame(
  Model = c("ARIMA(2,1,3)", "ARIMA(2,1,2)"),
  AIC = c(AIC(m1), AIC(m2))
)

# Create the formatted table using flextable
aic_table <- flextable(aic_results) %>%
  set_caption(caption = "Table 1: AIC Values for Different ARIMA Models") %>%
  theme_vanilla() %>%
  autofit() %>%
  add_footer_row(values = "Source: Self-prepared based on model outputs.",
                 colwidths = ncol(aic_results)) %>%
  width(j = 1, width = 2.5)

# Display the table
print(aic_table)

arima_results <- data.frame(
  Section = c(rep("Parameters", length(m1$coef)),
              rep("Model Metrics", 3)),
  Name = c(names(m1$coef),
           c("AIC", "BIC", "Log-Likelihood")),
  Value = c(round(m1$coef, 4),
            round(AIC(m1), 2),
            round(BIC(m1), 2),
            round(m1$loglik, 2)),
  Std_Error = c(round(sqrt(diag(m1$var.coef)), 4),
                rep(NA, 3))
)

# Create the formatted table using flextable
arima_table <- flextable(arima_results) %>%
  set_caption(caption = "Table 2: ARIMA (2,1,3) Model Results") %>%
  theme_vanilla() %>%
  autofit() %>%
  merge_v(j = "Section") %>%
  add_footer_row(values = "Source: Self-prepared based on ARIMA (2,1,3) model outputs.",
                 colwidths = ncol(arima_results))

# Display the table
print(arima_table)



#-----------------------
#TS2
#-----------------------

#Data
ts2 <- ts(data["ts2"])
plot(ts2)

# Differentiation
ts2.diff <- (diff(ts2,lag = 1))
plot(ts2.diff)

par(mfrow =c(2,2))
par(mfrow = c(2, 2), oma = c(3, 6, 3, 6), mar = c(3, 3, 4, 2))
acf(ts2,main = "Figure 7a: ACF TS2")
pacf(ts2,main = "Figure 7b: PACF TS2")
acf(ts2.diff,main = "Figure 7c: ACF TS2.Diff(1)")
pacf(ts2.diff,main = "Figure 7d: PACF TS2.Diff(1)")

```

```r
# Model and residuals
m1 <- arima(ts2, order = c(2, 0, 3))
plot(resid(m1),main="Figure Residual Time Series")
qqnorm(resid(m1)); qqline(resid(m1))
acf(resid(m1),main="ACF Residual"); pacf(resid(m1),main="PACF Residual")
mtext("Figure 8: Residual Analysis for ARIMA(2,0,3) Model", side = 3, line = 1, outer = TRUE, cex = 1)
m2 <- arima(ts2, order = c(2, 0, 2))
plot(resid(m2),main="Figure Residual Time Series")
qqnorm(resid(m2)); qqline(resid(m1))
acf(resid(m2),main="ACF Residual"); pacf(resid(m2),main="PACF Residual")
mtext("Figure 9: Residual Analysis for ARIMA(2,0,2) Model", side = 3, line = 1, outer = TRUE, cex = 1)

AIC(m1,m2)

aic_results <- data.frame(
  Model = c("ARIMA(2,0,3)", "ARIMA(2,0,2)"),
  AIC = c(AIC(m1), AIC(m2))
)

# Create the formatted table using flextable
aic_table <- flextable(aic_results) %>%
  set_caption(caption = "Table 3: AIC Values for Different ARIMA Models") %>%
  theme_vanilla() %>%
  autofit() %>%
  add_footer_row(values = "Source: Self-prepared based on model outputs.",
                 colwidths = ncol(aic_results)) %>%
  width(j = 1, width = 2.5)

# Display the table
print(aic_table)

```

```r
arima_results <- data.frame(
  Section = c(rep("Parameters", length(m1$coef)),
              rep("Model Metrics", 3)),
  Name = c(names(m1$coef),
           c("AIC", "BIC", "Log-Likelihood")),
  Value = c(round(m1$coef, 4),
            round(AIC(m1), 2),
            round(BIC(m1), 2),
            round(m1$loglik, 2)),
  Std_Error = c(round(sqrt(diag(m1$var.coef)), 4),
                rep(NA, 3))
)

# Create the formatted table using flextable
arima_table <- flextable(arima_results) %>%
  set_caption(caption = "Table 4: ARIMA (2,0,3) Model Results") %>%
  theme_vanilla() %>%
  autofit() %>%
  merge_v(j = "Section") %>%
  add_footer_row(values = "Source: Self-prepared based on ARIMA (2,0,3) model outputs.",
                 colwidths = ncol(arima_results))

# Display the table
print(arima_table)


```

```r
#-----------------------
#TS3
#-----------------------

#Data
ts3 <- ts(data["ts3"])
plot(ts3)
# Differentiation
ts3.diff <- (diff(ts3,lag = 1))
plot(ts3.diff)

par(mfrow = c(2, 2), oma = c(3, 6, 3, 6), mar = c(3, 3, 4, 2))
acf(ts3,main = "Figure 10a: ACF TS3")
pacf(ts3,main = "Figure 10b: PACF TS3")
acf(ts3.diff,main = "Figure 10c: ACF TS3.Diff(1)")
pacf(ts3.diff,main = "Figure 10d: PACF TS3.Diff(1)")

# Model and residuals
m1 <- arima(ts3, order = c(2, 1, 3))
plot(resid(m2),main="Figure Residual Time Series")
qqnorm(resid(m2)); qqline(resid(m1))
acf(resid(m2),main="ACF Residual"); pacf(resid(m2),main="PACF Residual")
mtext("Figure 11: Residual Analysis for ARIMA(2,1,3) Model", side = 3, line = 1, outer = TRUE, cex = 1)

m2 <- arima(ts3, order = c(2, 1, 2))
plot(resid(m2),main="Figure Residual Time Series")
qqnorm(resid(m2)); qqline(resid(m1))
acf(resid(m2),main="ACF Residual"); pacf(resid(m2),main="PACF Residual")
mtext("Figure 12: Residual Analysis for ARIMA(2,1,2) Model", side = 3, line = 1, outer = TRUE, cex = 1)

aic_results <- data.frame(
  Model = c("ARIMA(2,1,3)", "ARIMA(2,1,2)"),
  AIC = c(AIC(m1), AIC(m2))
)

```

```r
# Create the formatted table using flextable
aic_table <- flextable(aic_results) %>%
  set_caption(caption = "Table 5: AIC Values for Different ARIMA Models") %>%
  theme_vanilla() %>%
  autofit() %>%
  add_footer_row(values = "Source: Self-prepared based on model outputs.",
                 colwidths = ncol(aic_results)) %>%
  width(j = 1, width = 2.5)

# Display the table
print(aic_table)

arima_results <- data.frame(
  Section = c(rep("Parameters", length(m2$coef)),
              rep("Model Metrics", 3)),
  Name = c(names(m2$coef),
           c("AIC", "BIC", "Log-Likelihood")),
  Value = c(round(m2$coef, 4),
            round(AIC(m2), 2),
            round(BIC(m2), 2),
            round(m1$loglik, 2)),
  Std_Error = c(round(sqrt(diag(m2$var.coef)), 4),
                rep(NA, 3))
)

# Create the formatted table using flextable
arima_table <- flextable(arima_results) %>%
  set_caption(caption = "Table 6: ARIMA (2,1,2) Model Results") %>%
  theme_vanilla() %>%
  autofit() %>%
  merge_v(j = "Section") %>%
  add_footer_row(values = "Source: Self-prepared based on ARIMA (2,1,2) model outputs.",
                 colwidths = ncol(arima_results))

```

```r
# Display the table
print(arima_table)

# Plot Time Series
plot(ts1,main="Figure 3a: Time Series 1")
plot(ts2,main="Figure 3b:Time Series 2")
plot(ts3,main ="Figure 3c:Time Series 3")
```

4.3 R Code Question 3

```r
library(boot)
set.seed(123)
n <- 100
time.series <- ts(cumsum(rnorm(n)))
plot(time.series, main = "Original Time Series", ylab = "Value", xlab = "Time")
mean.function <- function(data, indices) {
  return(mean(data[indices]))
}
bootstrap1 <- boot(time.series, statistic = mean.function, R = 1000)
print(bootstrap1)
block.size <- 10
mean.function.tsboot <- function(data) {
  return(mean(data))
}
bootstrap2 <- tsboot(time.series, statistic = mean.function.tsboot, R = 1000,
                     l = block.size, sim = "fixed"
)
print(bootstrap2)
par(mfrow = c(1, 2))
hist(bootstrap1$t, main = "Bootstrap 1", xlab = "Mean", col = "lightblue", border = "black")
hist(bootstrap2$t, main = "Bootstrap 2", xlab = "Mean", col = "lightgreen", border = "black")
```